

Read ME File

Library Management System

Members:

William Selui (s11244878)

Sebastian Afeaki (s11206285)

Overview

This program is a simple Library Management System written in C++. It allows users to manage book data, view reports and update the number of copies for each book. The program reads book data from a text file (books.txt) and provides a menu-driven interface for various functionalities.

The program includes all the following functionalities:

1. Print all book details

- Displays the ID, title, number of copies available and the status (Available or Out of Stock) for each book.

2. Print total number of books in stock

- Calculates and displays the total number of copies of all books in the library

3. Print book with the highest number of copies available

- Finds and displays the book that has the maximum number of copies

4. Print percentage of books that out of stock

- Calculates and shows the percentage of books that currently have zero copies

5. Update the number of copies for a specific book

- Allows the user to update the number of copies for a book by entering its ID. Updates are saved back to the books.txt file

6. Exit

- Ends the program

Program Structure

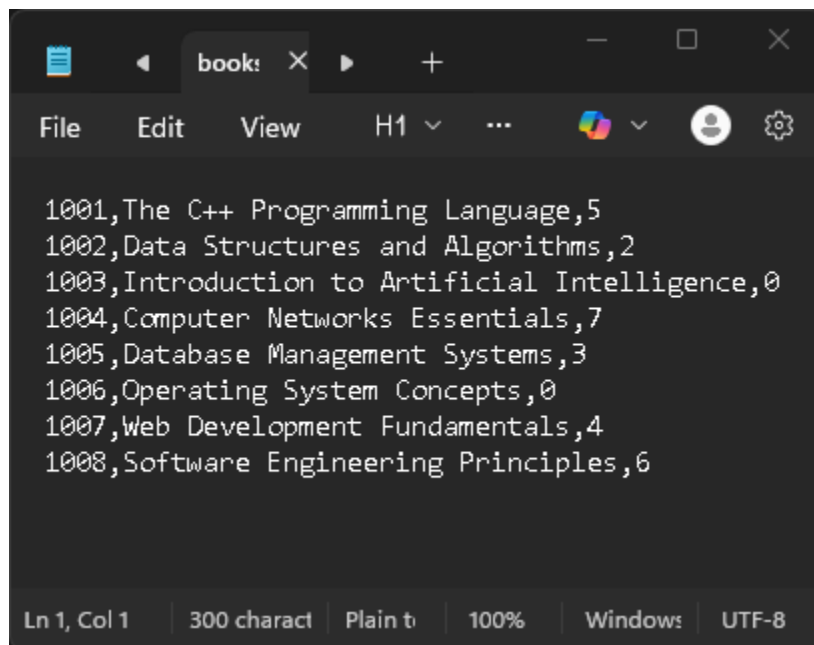
- **Struct Books:** Holds information for each book, including Book_ID, Book_Title and Copies-Available

Functions:

- **book_Status(int copies):** Returns "Available" if copies > 0, otherwise "Out of Stock".
- **print_Books_Details(Books books[], int size):** Displays all books with details.
- **print_Sum_of_copies(Books books[], int size):** Prints total number of copies across all books.

- **print_Book_with_highest_copies_available(Books books[], int size):** Prints the book with the most copies.
 - **print_percentage_books_out_of_stock(Books books[], int size):** Prints the percentage of books that are out of stock.
 - **update_functionality(Books books[], int size):** Updates the number of copies for a specific book and rewrites the file.
-
- **File Handling**
 - The program reads from books.txt at startup.
 - Any updates to book copies are written back to the same file.

Input File Format (books.txt)



The screenshot shows a code editor window with a dark theme. The title bar indicates the file is named 'book:'. The menu bar includes 'File', 'Edit', 'View', and 'H1'. The main text area contains the following data:

```
1001,The C++ Programming Language,5
1002,Data Structures and Algorithms,2
1003,Introduction to Artificial Intelligence,0
1004,Computer Networks Essentials,7
1005,Database Management Systems,3
1006,Operating System Concepts,0
1007,Web Development Fundamentals,4
1008,Software Engineering Principles,6
```

The status bar at the bottom shows 'Ln 1, Col 1', '300 charact', 'Plain t', '100%', 'Windows', and 'UTF-8'.

HOW TO RUN THE PROGRAM

1. Ensure you have a C++ compiler installed.
2. Place the books.txt file in the same directory as the program.
3. Compile the program
4. Run the program
5. Follow the on-screen menu options to use the system
 - Input numbers 1-6 to navigate the menu
 - When updating a book, enter the book ID and the new number of copies

NOTES

- Maximum number of books supported: 100 (Max_Input = 100)
- Input validation is included for menu choices and updating book copies
- The program will overwrite books.txt whenever a book's copies are updated

Mark Allocation Sheet

Group Name: SAIPE

PROJECT MANAGER: X (there was no project manager both work together)

Member ID	Percentage contribution of allocated task
S11244878	50%
S11206285	50%

Certification	I hereby certify that the information given is true and correct	
ID	Member Name	Signature
S11244878	William Selui	W.Selui
S11206285	Sebastian Afeaki	S.Afeaki